

SageMaker Deep Dive

Instructor

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Domains

Domain consists of

- A list of authorized users
 - An EFS volume
 - Private and public shared spaces
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- *All shared AWS SageMaker services can be seen by all users within a given domain*



Amazon SageMaker

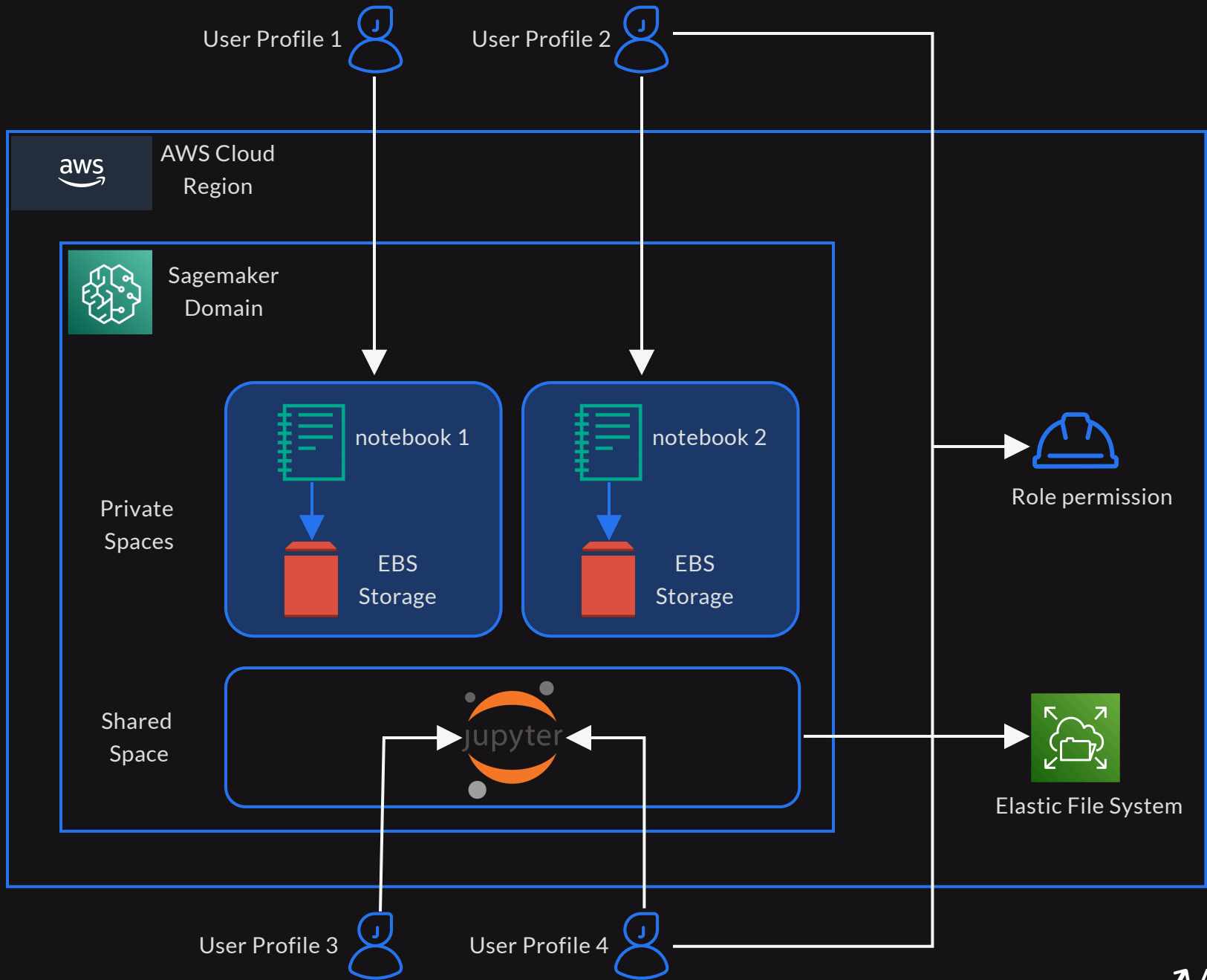
Domains

Domain can be created by:

- Quick setup
 - Domain created accessing the AWS resources using public Internet
- Standard setup
 - AWS resources are accessed internally without connecting to the internet
 - More fine-grained control over security and networking



Domains



SageMaker Virtual Machines

<https://aws.amazon.com/sagemaker/pricing/>

- EC2 servers are prefixed with ml for SageMaker
- Based on Amazon Linux 2 OS

On-Demand Pricing			
Studio Classic	JupyterLab	Code Editor	RStudio
	Notebook Instances	Processing	TensorBoard
Data Wrangler	Feature Store	Training	MLflow
	Real-Time Inference	Asynchronous Inference	
Batch Transform	Serverless Inference	JumpStart	Profiler
	HyperPod	Inference optimization	
Amazon SageMaker Studio Classic			
Studio Classic offers one-step Jupyter notebooks in our legacy IDE experience. The underlying compute resources are fully elastic and the notebooks can be easily shared with others, allowing seamless collaboration. You are charged for the instance type you choose, based on the duration of use.			
Region: Asia Pacific (Mumbai) ↕			
Standard Instances	vCPU	Memory	Price per Hour
ml.t3.medium	2	4 GiB	\$0.054
ml.t3.large	2	8 GiB	\$0.108
ml.t3.xlarge	4	16 GiB	\$0.215
ml.t3.2xlarge	8	32 GiB	\$0.43
ml.m5.large	2	8 GiB	\$0.121

SageMaker Virtual Machines

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Instance classes:

- Standard Instances: **t, m series**
- Compute optimized: **c series**
- Memory-optimized: **r series**
- GPU optimized: **g, p series**
- DL Inference optimized: **inf series**

On-Demand Pricing

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Data Wrangler	Feature Store	Training	MLflow	Real-Time Inference	Asynchronous Inference	
Batch Transform	Serverless Inference	JumpStart	Profiler	HyperPod	Inference optimization	

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Training with SageMaker

Built-in Algorithms

Default support for common algorithms from various providers

- **Default support for frameworks:**
 - PyTorch
 - Tensorflow
 - Hugging Face
 - Chainer
 - Scikit-learn
 - MXNet
 - Spark ML

Script Mode

- Container provided by AWS
- User provides script file to run with the container
- Support for training and deployment

Custom Container

- Create your deep learning container
- Support for training and deployment with other SageMaker resources

Thank You
